

FarmConnect

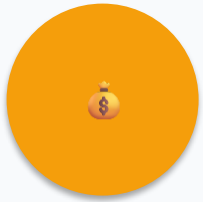
Empowering Indian Farmers with Local Resources & AI

Team: Tech-Vibe

A unified platform connecting farmers to equipment, labor & government schemes

Problem Statement

Why do Indian farmers struggle to access basic agricultural resources?



High Equipment Costs

86% of Indian farmers are small/marginal.
They cannot afford to buy tractors, harvesters, or tillers outright.



Labor Shortages

Finding reliable, skilled farm labor during peak seasons (sowing, harvest) is extremely difficult in rural areas.



Policy Information Gap

Hundreds of central & state schemes exist, but farmers are unaware of which ones they actually qualify for.



Language Barrier

Most agricultural platforms are English-only, excluding millions of regional language speakers.

Project Abstract

FarmConnect is a role-based web platform that bridges the gap between small/marginal farmers and the resources they need — farm equipment, skilled agricultural labor, and government welfare schemes.

The platform enables equipment vendors to list idle machinery for rent, agricultural workers to register their skills and availability, and farmers to discover these resources filtered by their city and specific needs. An AI-powered Policy Matcher, driven by Google's Gemini LLM, analyzes each farmer's crop type, land size, income, and state to recommend the most relevant government subsidies and credit schemes — eliminating the information gap that prevents millions from accessing financial support they are entitled to.

Built with Next.js, FastAPI, SQLite, and the Gemini API, FarmConnect features a glassmorphism UI, multi-language support for 9 Indian languages, and an intelligent search system — making it accessible to farmers across India.

3

User Roles

9

Languages

AI

Policy Engine

SQLite

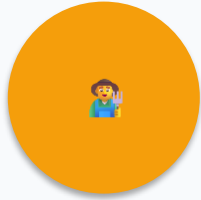
Database

Our Solution



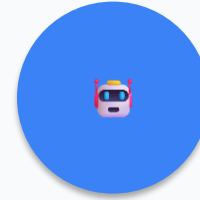
Equipment Rental

Vendors list idle tractors, harvesters, tillers & pumps. Farmers search by location and equipment type, then call vendors directly — zero middlemen.



Worker Discovery

Agricultural workers register their skills (harvesting, planting, spraying) and daily wage. Farmers filter by skill and city to find labor fast.



AI Policy Matcher

Powered by Google Gemini LLM. Farmers enter crop, land size, income & state. The AI returns personalized government schemes with how-to-apply steps.



Multi-Language

Entire platform translates instantly into Hindi, Malayalam, Tamil, Telugu, Kannada, Marathi, Gujarati & Bengali using integrated Google Translate.

Key Features

Intelligent Search Bar

The hero search bar intelligently routes queries:

- "harvester" → Equipment section
- "harvesting" → Workers section
- "loan" or "scheme" → AI Policy Matcher

Role-Based Access

Three distinct user roles:

- Farmer — discovers equipment & workers
- Vendor — lists equipment for rent
- Worker — registers skills & availability

Live Database Integration

All registrations (equipment & workers) are stored in a real SQLite database via FastAPI. Search results come from actual DB queries with location and type/skill filtering.

AI-Powered Scheme Matching

Gemini LLM analyzes farm profiles and returns 3-5 relevant central + state-specific schemes. Default schemes shown on page load; form acts as a filter for personalization.

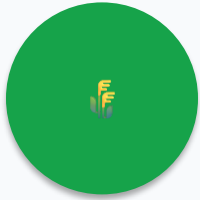
Glassmorphism UI Design

Premium modern interface with animated blobs, sliding sidebars, custom select dropdowns, and consistent micro-animations across every page and component.

9-Language Translation

Google Translate injected via a custom-styled dropdown in the navbar. Translates the entire app on-the-fly without breaking any React state or backend API connections.

User Roles & Journeys



Farmer

- ✓ Search for equipment by location & type
- ✓ Discover skilled workers for hire
- ✓ Use AI Policy Matcher for government schemes
- ✓ Call vendors/workers directly from results
- ✓ Switch language for regional accessibility



Equipment Vendor

- ✓ Register idle machinery (tractor, harvester, etc.)
- ✓ Specify rental price, condition & availability
- ✓ Get discovered by farmers in your district
- ✓ Receive direct phone calls from interested farmers
- ✓ Edit profile information anytime



Agricultural Worker

- ✓ Register specialized skills (harvesting, planting, etc.)
- ✓ Set daily wage and available date range
- ✓ Get discovered by farmers searching for labor
- ✓ Receive direct calls without middlemen
- ✓ Edit profile and update availability

Technology Stack



Frontend

Framework

Next.js 14 (App Router) + React

Styling

Custom Vanilla CSS (Glassmorphism Design System)

Components

Custom Select, Date Picker, Animated Blobs, Sidebar

Search

Intelligent keyword routing (NLP-lite)

i18n

Google Translate DOM injection (9 languages)

State

React Context API (RoleContext for auth)



Backend & AI

Framework

FastAPI (Python) with Uvicorn

Database

SQLite (Equipment & Workers tables)

AI Engine

Google Gemini API (gemini-flash models)

Fallback

Rule-based engine if LLM fails

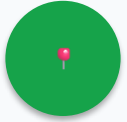
Caching

In-memory policy cache (reduces API costs)

API Design

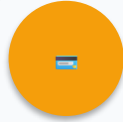
RESTful with v1 prefix + legacy aliases

Future Scope



GPS & Map Integration

Real-time distance calculation between farmers and vendors using geolocation. Map-based equipment tracking.



Integrated Payments

Secure escrow-based rental payments. UPI / wallet integration for seamless transactions.



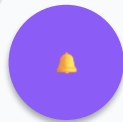
Mobile Application

React Native app for low-end Android devices, optimized for rural connectivity and offline-first capabilities.



Review & Rating System

Verified star ratings and reviews for vendors and workers, building trust and accountability.



Push Notifications

Alert farmers about seasonal schemes, nearby equipment availability, and new worker registrations.



Authentication & Security

OTP-based login via Aadhaar/phone, role-verified profiles, and encrypted data storage for privacy.

Conclusion

- ✓ FarmConnect eliminates middlemen — farmers connect directly with vendors and workers.
- ✓ AI-powered policy matching ensures no farmer misses a scheme they qualify for.
- ✓ Multi-language support makes the platform accessible to all of India's farmers.
- ✓ A real, working product — live database, live AI, live translations.

Thank You!

Team Tech-Vibe • Live Demo Available